

ASSIGNMENT #03

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
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Professor: Dr. Michael Caracotsios
Office: EIB 244

Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Name : Jacky Shen

Due Date: Friday, March 6th TA's Mailbox

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CHE 301 Homework 2

Jacky Shen

March 6th 2026

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Chapter 1

Problem A

Problem A: Thermal Properties of Nitrogen.

The following data on $\frac{P}{\rho RT}$ for Nitrogen gas were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

Calculate at temperature 10 °C and pressure 830 bar

1. The thermal expansion coefficient in **K⁻¹**
2. The pressure expansion coefficient in **K⁻¹**
3. The molar enthalpy in **kJ/mol**
4. The molar entropy in **kJ/(kmol K)**
5. The molar heat capacity C_p in **kJ/(kmol K)**
6. The molar heat capacity C_v in **kJ/(kmol K)**
7. The Joule-Thomson coefficient in **K/bar**
8. The speed of sound in **m/s**

The problem gave us a chart of Nitrogen Gas Compressibility factor in terms of Pressure in ATM and Temperature in degrees Celsius. To make all future calculations easier, I first did some interpolation with the following equations:

$$Z(10^{\circ}C) = Z(0^{\circ}C) + \frac{10 - 0}{25 - 0} [Z(25^{\circ}C) - Z(0^{\circ}C)] \quad (1.1)$$

$$Z(819.15 \text{ atm}) = Z(800 \text{ atm}) + \frac{819.15 - 800}{1000 - 800} [Z(1000 \text{ atm}) - Z(800 \text{ atm})] \quad (1.2)$$

	Temperature (C)					
P(atm)	0	10	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	0.999	1.000	1.000	1.001
5	0.998	0.998	0.999	1.000	1.001	1.001
10	0.996	0.997	0.998	1.000	1.001	1.002
30	0.959	0.974	0.996	1.001	1.005	1.007
50	0.958	0.973	0.996	1.004	1.009	1.014
100	0.984	0.992	1.004	1.018	1.027	1.033
200	1.036	1.044	1.057	1.072	1.082	1.089
300	1.132	1.138	1.146	1.154	1.160	1.163
400	1.256	1.255	1.254	1.253	1.250	1.247
600	1.524	1.512	1.495	1.471	1.450	1.432
800	1.798	1.768	1.723	1.697	1.658	1.625
819.15	1.824	1.794	1.748	1.719	1.678	1.644
1000	2.067	2.035	1.988	1.923	1.867	1.819

Since all of our calculations require the units to be in Kelvin and Bar, I then changed all the chart labeling to bar and Kelvin.

	Temperature (K)					
P(bar)	273.15	283.15	298.15	323.15	348.15	373.15
0	1.000	1.000	1.000	1.000	1.000	1.000
1.01325	1.000	1.000	1.000	1.000	1.000	1.000
3.03975	0.999	0.999	0.999	1.000	1.000	1.001
5.06625	0.998	0.998	0.999	1.000	1.001	1.001
10.1325	0.996	0.997	0.998	1.000	1.001	1.002
30.3975	0.959	0.974	0.996	1.001	1.005	1.007
50.6625	0.958	0.973	0.996	1.004	1.009	1.014
101.325	0.984	0.992	1.004	1.018	1.027	1.033
202.65	1.036	1.044	1.057	1.072	1.082	1.089
303.975	1.132	1.138	1.146	1.154	1.160	1.163
405.3	1.256	1.255	1.254	1.253	1.250	1.247
607.95	1.524	1.512	1.495	1.471	1.450	1.432
810.6	1.798	1.768	1.723	1.697	1.658	1.625
830	1.824	1.794	1.748	1.719	1.678	1.644
1013.25	2.067	2.035	1.988	1.923	1.867	1.819

Lastly, for preparation, it is much easier to calculate everything using density instead of the compressibility factor. Since the equation for compressibility factor was already given, I simply rearranged the equation to solve for density at 830 Bar and 283.15 Kelvin.

$$\rho = \frac{P}{ZRT}, \rho = 19.65741 (P = 830 \text{ Bar}, T = 10^\circ \text{C} = 283.15^\circ \text{K}) \quad (1.3)$$

With the density at the temperature and pressure of interest calculated, I can now move on to the actual calculations.

1.1 Thermal Expansion Coefficient

Thermal Expansion Coefficient, otherwise known as β , is calculated as follows:

$$\beta = -\frac{1}{\rho} \left(\frac{\partial \rho}{\partial T} \right)_P \quad (1.4)$$

The calculation requires the density and the density difference divided by the time difference at constant pressure. I already have the $\rho = 19.65741$ calculated, so I chose the constant pressure of 830 bar. For the two points of difference approximation, I chose $T = 298.15$ K and $T = 273.15$ K. The respective pressures calculated are 19.15139 and 20.04016. Plugging all the values into Equation 1.4, we get the following:

$$\beta = -\frac{1}{19.65741} \left(\frac{19.15139 - 20.04016}{298.15 - 273.15} \right) = 0.001809 K^{-1} \quad (1.5)$$

The Thermal Expansion Coefficient is

$$\boxed{1.809 \times 10^{-3} K^{-1}}$$

.

1.2 Pressure Expansion Coefficient

Pressure Expansion Coefficient, otherwise known as δ , is calculated as follows:

$$\frac{1}{P} \left(\frac{\partial P}{\partial T} \right)_\rho = - \frac{\left(\frac{\partial \rho}{\partial T} \right)_P}{\left(\frac{\partial \rho}{\partial P} \right)_T} \frac{1}{P} \quad (1.6)$$

The previous calculation already provided density at constant pressure, temperature at 0 degrees, and 25 degrees. Here, the calculation requires two additional densities. So I chose 10 degrees Celsius as the constant temperature and evaluated pressure at $P = 810.6$ Bar (800 ATM) and $P = 1013.25$ Bar (1000 ATM). The densities, respectively, are 19.47591 and 21.14659. Now with every value accounted for, plugging in all the numbers on the right side of equation .6.

$$- \frac{\left(\frac{19.15149 - 20.04016}{298.15 - 273.15} \right)_P}{\left(\frac{21.14659 - 19.47591}{1013.25 - 810.6} \right)_T} \frac{1}{830} = 0.005195 K^{-1} \quad (1.7)$$

The Pressure Expansion Coefficient is

$$\boxed{5.195 \times 10^{-3} K^{-1}}$$

.

1.3 Molar Enthalpy

In the CHE Calculator, N2 was added as component one. Since the enthalpy is wanted at 10°C , the command "HIG(comp1, 10) is used to get enthalpy at Ideal Gas, which is $-437.763 \frac{\text{J}}{\text{mol}}$. The Enthalpy calculation is as follows:

$$H = H^{ig} + H^{dep} \quad (1.8)$$

And the enthalpy departure is calculated as follows:

$$H^{dep} = RT \left[Z - 1 - T \left(\frac{\partial Z}{\partial T} \right)_P \right] \quad (1.9)$$

Plugging in values from the table and previous calculations, a H^{dep} value of $3878 \frac{\text{J}}{\text{mol}}$ is calculated. Plugging both values into equation 1.8:

$$H = -437.763 + 3878 = 3440.237 \frac{\text{J}}{\text{mol}} = 3.44 \frac{\text{kJ}}{\text{mol}} \quad (1.10)$$

The Molar Enthalpy is

$$\boxed{3.44 \frac{\text{kJ}}{\text{mol}}}$$

.

1.4 Molar Entropy

In the CHE Calculator, N2 was added as component one. Since the enthalpy is wanted at $10^{\circ}C$, the command "SIG(comp1, 10) is used to get entropy at Ideal Gas, which is $-57.2821 \frac{J}{mol}$. The Entropy calculation is as follows:

$$S = S^{ig} + S^{dep} \quad (1.11)$$

And the entropy departure is calculated as follows:

$$S^{dep} = -R \left[\ln Z + T \left(\frac{\partial Z}{\partial T} \right)_P \right] \quad (1.12)$$

Plugging in values from the table and previous calculations, a S^{dep} value of $2.25 \frac{J}{molK}$ is calculated. Plugging both values into equation 1.11:

$$S = -57.2821 + 2.25 = -55.0321 \frac{kJ}{kmolK} \quad (1.13)$$

The Molar Enthalpy is

$$\boxed{-55.0321 \frac{kJ}{kmolK}}$$

.

1.5 Molar Heat Capacity C_p

Similar to both molar enthalpy and molar entropy, C_p can be calculated with ideal gas + departure. Using the command "CPIG" in the CHE Calculator, an ideal value of $29.19192 \frac{kJ}{kmolK}$ is found. Here is how to calculate departure:

$$C_p^{dep} = -RT \left[2 \left(\frac{\partial Z}{\partial T} \right)_P + T \left(\frac{\partial^2 Z}{\partial T^2} \right)_P \right] \quad (1.14)$$

Plugging in values from the table, an answer for departure is calculated to be $14.2 \frac{kJ}{kmolK}$. Adding the two values together:

$$C_p = 29.19192 \frac{kJ}{kmolK} + 14.2 \frac{kJ}{kmolK} = 43.39 \frac{kJ}{kmolK} \quad (1.15)$$

The Molar Heat Capacity C_p is

$$\boxed{43.39 \frac{kJ}{kmolK}}$$

.

1.6 Molar Heat Capacity C_v

Molar Heat Capacity C_v is a part of this equation:

$$C_p - C_v = \frac{T\beta^2}{\rho\zeta} \quad (1.16)$$

Plugging all the numbers calculated so far:

$$C_p - C_v = \frac{(283.15)(0.001809)^2}{(19.71)(0.005195)} = 0.00905 \text{ L} \cdot \text{atm mol}^{-1} \text{K}^{-1} \quad (1.17)$$

$$C_v = 43.39 - 0.917 = 42.47 \text{ J mol}^{-1} \text{K}^{-1} \quad (1.18)$$

The Molar Heat Capacity C_v is

$42.47 \frac{\text{kJ}}{\text{kmolK}}$
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1.7 Joule-Thomson Coefficient

Joule-Thomson Coefficient μ_{JT} can be calculated like this:

$$\mu_{JT} = \frac{\beta T - 1}{\rho C_p} \quad (1.19)$$

Plugging in values previously calculated

$$C_p = \frac{43.39}{101.325} = 0.4283 \text{ L} \cdot \text{atm mol}^{-1} \text{K}^{-1} \quad (1.20)$$

$$\mu_{JT} = \frac{(0.001809)(283.15) - 1}{(19.71)(0.4283)} = \mu_{JT} = -0.0578 \text{ K/atm} = \mu_{JT} = -0.0570 \text{ K/bar} \quad (1.21)$$

The Joule-Thomson Coefficient is

$-0.0570 \frac{\text{K}}{\text{bar}}$

.

1.8 Speed of Sound

The speed of sound equation is as follows:

$$v^2 = \frac{C_p}{C_v} \left(\frac{\partial P}{\partial T} \right)_\rho \quad (1.22)$$

The equation can be split into two parts: C_p/C_v and the differential. The differential was already solved in the earlier step of pressure expansion coefficient, so I use the same number: 4.31226. C_p/C_v is 43.39/42.47, and everything together is roughly 4.4. Here we convert bar to pascal, multiply by 10^5 , and take square root.

$$v^2 = \left(\frac{43.39}{42.47} (4.4) \right)^5 = 440567.3685 v = 663.75 \frac{m}{s} \quad (1.23)$$

The Speed of Sound is

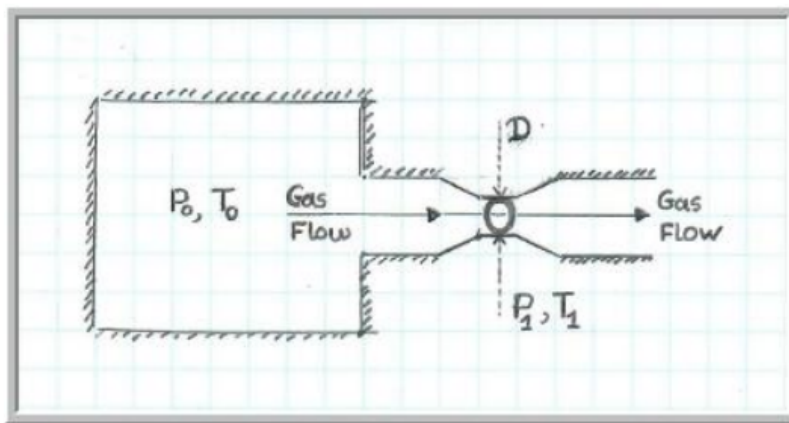
$$\boxed{663.75 \frac{m}{s}}$$

Chapter 2

Problem B

Problem B: Ideal Gas Venting.

Consider the venting of ethylene gas from the carbon steel tank shown in the diagram below



As the pressure begins to drop the gas velocity increases until it reaches the speed of sound in the narrow section of the pipe. **This is known as the critical condition.** The initial pressure and temperature in the tank are equal to 200 bar and 400 K respectively. The volume of the tank is equal to 20 m³

Calculate:

1. The initial mass in **kg** of ethylene in the tank
2. The temperature and pressure at the narrow section of the pipe when the critical mass flow rate which corresponds to the speed of sound (critical condition) is $m=5000$ kg/hr.
3. The diameter in **mm** of the narrow section of the venting pipe
4. The mass in **kg** of ethylene in the tank when the critical condition is reached
5. When during the venting process the pressure gauge reads $P_1=2300$ **psig** what is the mass in **kg** of ethylene in the tank

Assumptions and Hints

1. **Assumption #1:** The venting process is **reversible** and **adiabatic**
2. **Assumption #2:** Ethylene behaves as an ideal gas
3. **Hint #1:** You must start with the unsteady-state mass, energy and entropy balances
4. **Hint #2:** The kinetic energy of the gas at the narrow section of the pipe cannot be ignored.

2.1 Initial Mass

First, the ethylene gas is assumed to be an ideal gas. Which makes the ideal gas equation 2.1 valid.

$$PV = nRT \quad (2.1)$$

Rearranging this equation, as well as considering $n = \frac{m}{MW}$, I get equation 2.2.

$$m = \frac{PVMW}{RT} \quad (2.2)$$

Plugging in $200 \text{ bar} = 2 * 10^7 \text{ Pa}$, 20 m^3 , $0.02805 \frac{\text{g}}{\text{mol}}$, 8.314 and 400 K , I get:

$$m = \frac{(20000000)(20)(0.02805)}{(8.314)(400)} = 3373.8 \text{ kg} \quad (2.3)$$

The Initial Mass in kg of ethylene in the tank is

$$\boxed{3373.8 \text{ kg}}$$

2.2 Temperature and Pressure

To find temperature and pressure, first, I used CPIG and CVIG in the CHE Calculator to find C_p and C_v of ethylene gas in ideal gas conditions. It is 54.20603 and 45.89157, respectively. Then, the ratio is found.

$$\gamma = \frac{C_p}{C_v} = \frac{54.2}{45.89} = 1.18 \quad (2.4)$$

These two equations will be used to calculate pressure and temperature at the narrow section.

$$\frac{P^*}{P_0} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}} \quad (2.5)$$

$$\frac{T^*}{T_0} = \frac{2}{\gamma + 1} \quad (2.6)$$

Plugging in known values and γ :

$$T^* = 400 \left(\frac{2}{1.18 + 1} \right) = 366.97K \quad (2.7)$$

$$P^* = 200 \left(\frac{2}{1.18 + 1} \right)^{\frac{1.18}{1.18 - 1}} = 113.68bar \quad (2.8)$$

The Temperature and Pressure at Choke are, respectively,

$366.97K, 113.68bar$

2.3 Diameter

First, convert the mass flow rate from $\frac{kg}{hr}$ to $\frac{kg}{s}$.

$$\dot{m} = 5000 \text{ kg/hr} = \frac{5000}{3600} = 1.3889 \text{ kg/s} \quad (2.9)$$

Second, find the specific gas constant for ethylene.

$$R_s = \frac{R}{MW} = \frac{8.314}{0.02805} = 296.40 \text{ J/(kg} \cdot \text{K)} \quad (2.10)$$

Third, using question 2 throat conditions, convert $P = 113.68 \text{ bar} = 113.68 \times 10^5 Pa$.

Fourth, find density, velocity, and use them to find area, with area, find the diameter.

$$\rho^* = \frac{P^*}{R_s T^*} = \frac{113.6 \times 10^5}{(296.4)(366.97)} = 104.44 \text{ kg/m}^3 \quad (2.11)$$

$$v^* = \sqrt{\gamma R_s T^*} = \sqrt{(1.18)(296.4)(366.97)} = 358.26 \text{ m/s} \quad (2.12)$$

$$A^* = \frac{\dot{m}}{\rho^* v^*} = \frac{1.3889}{(104.44)(358.26)} = 3.71 \times 10^{-5} \text{ m}^2 \quad (2.13)$$

$$D = \sqrt{\frac{4A^*}{\pi}} = \sqrt{\frac{4(3.71 \times 10^{-5})}{\pi}} = 6.87 \times 10^{-3} \text{ m} = 6.87 \text{ mm} \quad (2.14)$$

The Diameter in mm is

$$\boxed{6.87 \text{ mm}}$$

2.4 Mass Left in Tank

First, I found the ratio between the pressure at the critical point and the pressure at the inlet:

$$\frac{P^*}{P} = \frac{113.68}{200} = 0.5684 \quad (2.15)$$

Then I relate this to the surroundings. Since the gas is venting to the atmosphere, $P = P_{atm} = 1.013$ bar. $1.013/0.5684 = 1.78$ bar absolute.

$$\frac{T}{T_i} = \left(\frac{P}{P_i} \right)^{\frac{\gamma-1}{\gamma}} \quad (2.16)$$

$$T = 400 \left(\frac{1.78}{200} \right)^{\frac{1.181-1}{1.181}} = 194K \quad (2.17)$$

Then, I plug everything into equation 2.18 to calculate the mass.

$$m = \frac{PVMW}{RT} = \frac{(1.78 \times 10^5)(20)(0.02805)}{(8.314)(194)} = 61.8kg \quad (2.18)$$

The Mass Left in Tank is

$$\boxed{61.8kg}$$

2.5 Mass in Tank During Venting

First, convert the gauge pressure to absolute pressure. Using Atmospheric pressure = 14.7 psi.

$$P_{abs} = P_g + P_{atm} = 2300 + 14.7 = 2314.7 \text{ psia} = 159.59 \text{ bar absolute} \quad (2.19)$$

Second, find the temperature at the pressure. Here, I used the same equation used in question 2.4.

$$\frac{T}{T_i} = \left(\frac{P}{P_i} \right)^{\frac{\gamma-1}{\gamma}} \quad (2.20)$$

$$T = 400 \left(\frac{159.59}{200} \right)^{\frac{1.181-1}{1.181}} = 386.4K \quad (2.21)$$

Lastly, find mass with the same equation as question 2.4.

$$m = \frac{PVMW}{RT} = \frac{(1.5959 \times 10^7)(20)(0.02805)}{(8.314)(386.4)} = 2787kg \quad (2.22)$$

The Mass in kg of ethylene in the tank is

$$\boxed{2787kg}$$

Chapter 3

Conclusion

Problem A	
Thermal Expansion Coefficient	$1.809 \times 10^{-3} \text{ K}^{-1}$
Pressure Expansion Coefficient	$5.195 \times 10^{-3} \text{ K}^{-1}$
Molar Enthalpy	$3.44 \frac{\text{kJ}}{\text{mol}}$
Molar Entropy	$-55.0321 \frac{\text{kJ}}{\text{kmolK}}$
Molar Heat Capacity C_p	$43.39 \frac{\text{kJ}}{\text{kmolK}}$
Molar Heat Capacity C_v	$42.47 \frac{\text{kJ}}{\text{kmolK}}$
Joule-Thomson Coefficient	$-0.0570 \frac{\text{K}}{\text{bar}}$
Speed of Sound	$663.75 \frac{\text{m}}{\text{s}}$

Problem B	
Initial Mass	3373.8 kg
Temperature	366.97 K
Pressure	113.68 bar
Diameter	6.87 mm
Mass Left in Tank	61.8 kg
Mass in Tank During Venting C_p	2787 kg

ChatGPT was used for some equation development and code generation. CHE Calculator was used for any and all ideal gas data, including C_p , C_v , etc.